BPH

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 148	Anatomy & Development of the Prostate
Ch 149	BPH: Aetiology, Pathophysiology & Natural History
Ch 150	Evaluation & Nonsurgical Management of BPH
Ch 151	Minimally Invasive & Endoscopic Management of BPH
Ch 152	Simple Prostatectomy
★	Putting it together: the BPH pathway

148 Anatomy and Development of the Prostate

1 · Development (embryology)

- **Origin**: urogenital sinus (UGS; endoderm in mesoderm). Female UGS forms prostate with enough androgen. Incomplete cloacal division (imperforate anus, urorectal fistula) → impaired development.
- **DHT**: †5α-reductase type 2 in UGS mesenchyme converts testosterone → DHT (more potent AR ligand). **SRD5A2** loss → incomplete prostate; castration or inactive AR → absent. †Same enzyme is the 5-ARI target in BPH.
- **Androgen not rate-limiting**: testosterone present week 9, budding week 10. Gate = **AR demethylation** in UGS (epigenetic checkpoint).

Week	Event
6–7	Cloaca divides → ventral UGS, dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR → andromedins instruct epithelium
10	Epithelial buds invade mesenchyme; fix adult ductal pattern

Mesenchyme instructs, epithelium responds

- **Lasnitzki–Mizuno** recombination: buds need **mesenchymal** AR; epithelial AR not needed for initiation.
- **Cunha**: UGS mesenchyme reprogrammed adult bladder epithelium into prostate → **instructive** induction.
- Epithelial specification: **Nkx3-1**, Edar, Wnt10b.
- **Andromedins**: androgen-induced stromal signals — **FGF10** (required), FGF7, Wif1, Wnt2; FGF10 and retinoic acid major roles.
- Branching needs **stromal** AR; Sonic hedgehog = epithelial paracrine factor. Oestrogens not needed for budding.

> **KEY POINT**
> McNeal: **re-awakening of embryonic induction** in ageing TZ stroma may initiate new glands → BPH.

- **Environmental**: gestational DES → squamous metaplasia; phthalates, TCDD → agenesis/epimutations. TCDD → **hyperinnervation** → hyperactive smooth muscle, frequency — may explain variable α-blocker response.

2 · Zonal anatomy (McNeal)

- P1 Anatomy of prostate and neurovascular bundles: clinical and surgical concepts in localised prostate cancer — *zones (PZ 70% glands/70% cancers, CZ 25%, TZ 5%/BPH, AFMS), 35° urethral bend, surgical capsule, fasciae, NVB at 5/7 o'clock, urethral/capsular arteries, Santorini plexus, lymph nodes*
- Urethra bends **≈35°** at verumontanum; **13–15** ducts open dorsolaterally at the verumontanum — distal limit of TURP/enucleation.
- TZ: shortest ducts, †≈20% of cancers; PZ: longest ducts.

KEY POINT

Trap: CZ (around ejaculatory ducts) ≠ TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). Enlarging TZ compresses PZ into the **surgical (pseudo)capsule** — the enucleation plane → *Ch 152*.

3 · Fascia, vessels †

- †Capsule incomplete anteriorly/apically. NVBs lie outside the capsule → spared by enucleation.
- †Prostatic veins communicate with vertebral (**Batson**) plexus → spinal metastases.

4 · Histology and cell types

- Acinus: basal + luminal epithelium; stroma of fibroblasts + smooth muscle.

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	PSA, NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, PAP	Almost none

- **Urethra + proximal ducts**: lung-like epithelium — **club** (SCGB1A1), **hillock** (KRT13), neuroendocrine (CHGA; mostly neural crest–derived). Handover to NKX3-1+ epithelium = cellular "transition zone".
- BPH stroma: prominent T, B, dendritic cells.

5 · Secretions and clinical markers

- Prostatic fluid: **37–44%** of semen, acidic (pH ≈6.3); seminal vesicle fluid (majority, alkaline) supplies semenogelins.
- Liquefaction by kallikreins within **20–30 min**; nonliquefaction if incomplete by 60 min.
- **Ion trapping**: macrolides concentrate in acidic fluid; inflammation alkalinises.
- PSA, PSMA, PAP are all extraprostatic ("prostate-specific" misnomer). Mouse has no PSA gene.

Product	Key facts
PSA (KLK3)	Serine protease (19q13.4); liquefies semen
PSMA	Folate hydrolase; rarely deleted in cancer → PSMA-PET, ^{177}Lu-PSMA
PAP	First tumour marker MCQ (Huggins and Hodges, 1941)
Zinc	Highest of any tissue; inhibits mitochondrial aconitase → citrate secreted

KEY POINT

Benign: high zinc blocks aconitase → citrate exported. Cancer: **zinc lost**, citrate oxidised → low citrate (†basis of MR spectroscopy).

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

1 · Definitions and the four overlapping entities

- P2 Pathophysiology, classification and management of LUTS. — *BPH vs BPE vs BPO vs BOO, ICS symptom phases, static/dynamic (α1A) components, capsule/TUIP, bladder hypertrophy → decompensation, one-third keep storage LUTS after surgery*

Term	How measured
BPH — histological ↑ epithelial + stromal cell number in periurethral/TZ	Histology
BPE — enlargement from BPH	DRE, TRUS, MRI
BOO — any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow (best); flow rate (indirect)
BPO — BOO caused by BPE	Pressure–flow
LUTS — storage, voiding, post-micturition; not prostate-specific	IPSS

- The four circles overlap only **partially**; BOO may exist without LUTS, BPE or BPH (e.g. stricture).
- Non-prostatic LUTS: detrusor ageing, polyuria, sleep disorders, neurogenic/primary bladder disease. Impaired contractility mimics obstruction.
- "**Prostatism**" obsolete; Paul Abrams coined "LUTS" MCQ. "Hypertrophy" wrong: ↑ cell **number**.
- Storage symptoms bother more than voiding; post-micturition dribble commonest. Age-matched women report similar LUTS.
- †BOOI = PdetQmax − 2Qmax: **>40 obstructed, 20–40 equivocal, <20 unobstructed**.

> **KEY POINT**
> Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are separate; obstruction is diagnosed only by pressure–flow.

2 · Epidemiology

- Autopsy: **no BPH under 30**; ≈ **50% in the 50s**†, **88% in the 80s**. **Similar across all ethnic/geographic groups**.
- Clinical prevalence depends on definition (none globally accepted); prevalence matters more than incidence.
- Three axes: **symptoms** (IPSS), **enlargement**, **obstruction** (+ bother, QoL). Proposed volume cut-off **>30 mL**; community composite IPSS >7, Qmax <15 mL/s, volume >20 mL.
- P3 P1 IPSS score; Methods to estimate prostate size — *IPSS items, bands 0–7/8–19/20–35, QoL item, no incontinence/pain items, LURN; DRE underestimates, TRUS ellipsoid, MRI, CT*
- SF-36 barely affected by LUTS → need disease-specific QoL tools. DRE underestimates by 25% to ≥50%; MRI reads ≈10% larger than TRUS.
- TPV ≈ **25 mL in the 30s → 35-45 mL in the 70s**; TZV ≈ 15 → 25 mL.
- Qmax falls almost linearly with age (Olmsted median 20.3 → 11.3 mL/s, 40–44 vs 75–79). Obstruction probability: Qmax <10 high, 10–15 intermediate, >15 low. Qmax depends on

voided volume, varies day to day; penile cuff test approximates pressure–flow.

3 · Aetiology

KEY POINT
BPH is an **accumulation** disease: volume = proliferation − death. Established BPH: equal/↓ replication, ↑ bcl-2; androgens act chiefly by **suppressing cell death** → withdrawal shrinks gland. Early BPH may involve rapid proliferation.

- New gland formation otherwise only fetal → **embryonic reawakening** of stromal induction.

3.1 Androgens — permissive, not causal

- Prepubertal castrates/androgen defects **never develop BPH**; withdrawal → **partial** involution. Two established determinants: **age** and **functioning testes**.
- Serum testosterone falls with age but **intraprostatic DHT and AR stay high**; DHT **maintained, not elevated** (old "raised DHT" used cadaveric controls) MCQ.
- Serum testosterone does not predict volume or PSA (only BMI tracks low testosterone); higher baseline serum DHT predicted 20-year BPH risk.
- Prostate keeps AR for life; penile AR lost after puberty. T/DHT not direct mitogens — act via growth factors. Androgens are **anti-inflammatory**. AR CAG/vitamin D receptor polymorphisms: minor.

ALGORITHM
Testosterone → stromal cell → **type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) → DHT → autocrine on stroma or paracrine to epithelium → DHT–AR → DNA → growth factors → epithelial survival/growth

- ≈ **90%** of prostatic androgen is testicular DHT; ≈10% adrenal. DHT: higher AR affinity, more stable complex.
- Withdrawal: switches off androgen-dependent genes (PSA), activates cell death and inflammatory genes.

Feature	Type 1	Type 2
Site	Skin, liver	Prostate **stromal fibroblasts**, some basal cells
5α-reductase deficiency	Normal	**Mutated**
Finasteride / dutasteride	— / inhibited	Inhibited / inhibited
Role	Possibly cancer	Development and BPH growth

- Finasteride shrinks gland as much as dual inhibition (≈castration) → type 1 unimportant in BPH.
- SRD5A2 on **chromosome 2p23** MCQ: V89L ↓, A49T ↑ activity; BPH links inconsistent. Promoter methylation in up to 30% of BPH.

3.2 5-ARI resistance

- 5-ARIs: **only medical therapy that reduces volume**.
- Resistance is **not** failure to lower DHT: focal lack of luminal atrophy in nodules, or stromal hyperplasia. Mechanisms (inflammation-linked): club-cell transition with high NF-κB (→ AR variants); SRD5A2 silencing → oestrogenic survival.
- Proposed definition: volume unchanged + no atrophy + DHT decreased.

- On 5-ARI/ADT, luminal cells become club-like (lineage plasticity).

3.3 Oestrogens

- Dog: androgen + oestradiol → BPH; oestrogen induces AR, sensitises gland, ↑ stroma/collagen; both inhibit cell death.
- Oestrogen rises with age (relative to testosterone); larger glands ↔ higher oestradiol; adipose aromatisation.
- **ERα = stroma** (proliferation, ↑ FGF-2/FGF-7); **ERβ = epithelium** (antiproliferative, anti-inflammatory). ERβ agonist failed in a LUTS RCT.

3.4 Cell death; stromal–epithelial interaction

- Apoptosis is active (needs RNA/protein synthesis), no immune activation; after castration TGF-β is the likely mediator.
- ECM regulates epithelial phenotype (epithelium on collagen stays differentiated).
- Hypothesis: **stromal defect removes a brake** on epithelial proliferation. Periurethral interstitial fibroblasts (type 2 5α-reductase) ↑ in BPH, expand TZ; **no drug targets them**.
- CXC chemokines/IL-8 → myofibroblasts → collagen, fibrosis → LUT dysfunction.

3.5 Growth factors

Androgen permits; growth factors signal (likelier primary defect). First prostatic growth factor: basic FGF (FGF-2) MCQ.

Factor	Source → target	Role in BPH
FGF-2	Stroma (autocrine)	↑ 2–3-fold; induced by TGF-β, hypoxia
FGF-7 (KGF)	Stroma → receptor **only on epithelium**	Leading stromal mediator of hormonal control
FGF-10	Smooth muscle → epithelium	↑ by androgens
TGF-β1	Stroma	**Inhibits epithelium**, **mitogenic for fibroblasts**; BPH stroma blunted response
IGF-1, -2	—	Transgenic IGF-1 → enlargement; serum IGF-1 unrelated to volume
Others	EGF, VEGF, FGF-17	

- **Hypoxia hypothesis**: hypoxia → HIF-1 → FGF-2/FGF-7 → nodule growth → more hypoxia; vascular disease (diabetes) may be a risk factor.
- α-blockade induces apoptosis in some models.

3.6 Inflammation

- **Activated T cells** secrete VEGF, HB-EGF, FGF-2; IL-2, IL-7, IFN-γ drive stroma; senescent epithelium secretes IL-8; COX-2 ↑.
- Autoimmune disease ↑ BPH; TNF antagonists ↔ ↓ BPH incidence.
- **Inflammation correlates with enlargement and 5-ARI resistance**; causality unproven.

3.7 Metabolic syndrome, lifestyle

- Metabolic syndrome → prostatic inflammation → LUTS/BPH. Mechanisms: aromatisation; **hyperinsulinaemia → sympathetic overactivity**; low HDL.
- **Waist circumference** predicts prostatectomy/LUTS independent of BMI. BMI >35 vs <25: OR **3.5** for prostate ≥40 mL. NAFLD associated.

Factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (LUTS)	↑ risk
Physical activity; moderate alcohol	↓ risk (alcohol ↓ BPH, **not LUTS**)
Smoking, diet, vasectomy, sexual activity	No consistent effect (**vasectomy not a risk factor**)
Jewish religion	↑ prostatectomy, not BPH (access) MCQ

- **Depression** strongly associated with LUTS; LUTS severity correlates with erectile/ejaculatory dysfunction.
- **Drugs worsening LUTS**: α-sympathomimetic cold remedies; antidepressants, antihistamines, bronchodilators (+2–3 IPSS); anticholinergics/adrenergics → retention.

3.8 Genetics and others

- Young men with large glands (>37 g): first-degree relatives ≈ **4-fold** risk of surgery (brothers ≈6–8-fold). **Autosomal dominant**; ≈ **50%** operated before 60 heritable vs ≈9% after 60.
- **Familial BPH**: early onset, **large gland**, normal androgens and 5-ARI response. BPH and cancer genetic risks correlate.
- Only **dogs and humans** get BPH; seminal vesicle never hyperplastic. Testicular non-androgenic factors may contribute.

4 · Pathology

- **Every nodule arises in the TZ or periurethral region**. TZ: two glands outside preprostatic sphincter, ducts at urethral angulation near verumontanum. Periurethral zone: inside sphincter, glands parallel to urethra.
- **Middle lobe** is **periurethral** (no TZ at bladder neck) MCQ. Patterns: middle, lateral, trilobar, posterior commissural (median bar).

> **KEY POINT**
> Human-only **capsule** transmits expansion onto the urethra; dog gets BPH but rarely obstructs (no capsule). TUIP relieves obstruction without volume change. Hence **size does not correlate with obstruction**.

Feature	Periurethral nodules	TZ nodules
Composition	Purely stromal (early), like **embryonic mesenchyme**	Glandular, tall epithelium
Origin	Interstitial fibroblasts	New duct branches → new ductal system

- **Two phases**: first ≈20 years ↑ nodule number; then growth of (glandular) nodules.
- Small resected glands mainly stroma; large enucleated glands mainly epithelial nodules. Variable stromal:epithelial ratio → variable drug response.
- In symptomatic BPH, smooth muscle + connective tissue exceed epithelium. α-blockers target smooth muscle; 5-ARIs epithelium.

5 · Pathophysiology of obstruction

ALGORITHM

Hyperplasia → ↑ urethral resistance (static + dynamic) → detrusor compensation → cost in storage function → storage LUTS ± impaired emptying → (minority) decompensation

- Static: bulk compressed by capsule; ECM elastic elements; **not reduced** by α-blockade. Dynamic: **α1a** (most abundant).
- BPH smooth muscle **dedifferentiated** (↓ SM myosin heavy chain). Stromal cell fall after ablation may take ≥1 year.
- Other regulators: endothelin, kallikrein–kinin, **PDE4/5** → PDE5 inhibitors help LUTS. Autonomic overactivity correlates with symptoms.
- Many LUTS arise from **obstruction-induced bladder change**. ≈ **One-third** keep voiding dysfunction (mostly storage) after relief.
- Storage change: DO, ↓ compliance. Emptying change: ↓ contractility → detrusor failure (minority).

ALGORITHM

Three-stage remodelling: smooth muscle hypertrophy → compensation → decompensation (underactive bladder)

- Stimuli: stretch, pressure, **hypoxia/ischaemia**. Detrusor switches to **secretory phenotype** (ECM), mitochondrial and calcium-signalling dysfunction.
- **Trabeculation** = ↑ detrusor **collagen**; severe → significant residual; no accepted grading. Sequence: trabeculation → cellules → diverticula → failure. Ageing alone reproduces some changes.

KEY POINT

Storage LUTS persist after TURP because the collagen-rich, overactive detrusor does not reverse — counsel men with DO preoperatively. AUR is **not** the inevitable end: many in AUR have adequate detrusor and a precipitant.

6 · Natural history

- Best data: longitudinal population studies (no selection).
- **Regression to the mean**: entry thresholds select temporary extremes → spurious improvement; stricter threshold, larger effect. Part of every placebo effect.
- **Restriction of range**: in trials IPSS, Qmax and volume barely correlate; only TPV–TZV and PSA–volume survive.
- Outcomes: **direct** (symptoms, AUR, surgery) vs **proxy** (Qmax, residual, volume).
- ≈ **40%** improve without treatment (watchful waiting = placebo); active drug ≈ twice placebo improvement in each stratum.
- Mild LUTS: ≈1 in 3 progress over 4 years. Community: regression as common as progression.
- **Olmsted County**: IPSS ≈ **0.2–0.34 points/year** (steepest in 60s); Qmax ≈ **2.1%/year** fall (rapid if age ≥70, Qmax <10); volume ≈ **0.6 mL/year** (≈1.9%/year, faster with higher PSA/volume). Volume >50 mL ≈3.5-fold odds of moderate–severe LUTS.
- PSA–volume log-linear but imprecise in individuals. Perceptible improvement ≈ −2 points at low vs −6 at high baseline.

Predictor of progression	Predicts
Age	Everything; AUR rises linearly 40 → 80+
PSA ≥1.4–1.5 ng/mL	Growth, symptoms/flow, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, incident LUTS
IPSS ≥8 / high bother	AUR, surgery
Qmax <10–12 mL/s	AUR, flow decline, treatment
Raised PVR	Treatment failure, AUR

- **MTOPS** placebo: **4.5 per 100 patient-years** (17% at 4 yr); mostly **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.
- **PLESS** placebo: drift back to baseline, fastest in top PSA tertile (3.3–10); surgery 10%/4 yr, ≈6% → ≈15% across PSA/volume tertiles.
- **REDUCE** placebo: volume 40–80 mL → HR 1.67 incident LUTS; Qmax <10 → 1.67.

> **KEY POINT**
> PSA = cheap surrogate for volume, the main driver of AUR and surgery; >100-predictor models barely beat PSA alone.

7 · Complications

> **KEY POINT**
> Death, renal failure, stones, infection are now rare. Common: **AUR and surgery**, predicted by age, symptoms, low flow and large gland (PSA).

7.1 Acute urinary retention

- Historically 25–33% of TURP indications. Precipitants: infection, overdistension, fluid, alcohol, sex, bed rest; prostatic infarction **not** proven.

	Spontaneous AUR	Precipitated AUR
Trigger	None	Surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- Incidence: community ≈ **4.5–6.8/1000 person-years**; VA watchful waiting 9.6; PLESS placebo 18; MTOPS 6. Constant hazard.
- **Age × symptoms multiply**: Olmsted 70s with IPSS >7 → 34.7/1000 person-years. RR: IPSS ≥8 3.2; Qmax <12 3.9; volume >30 mL 3.0; age ≥70 up to ≈15.
- PSA >1.4 → several-fold; volume >40 mL ≈3-fold. Best IPSS items: incomplete emptying, frequency, weak stream. Recurrence: older age, diabetes, neurological disease.

7.2 Other complications

- **Renal**: older series ≈1 in 7 at TURP had renal failure (complications/mortality up to 6-fold). "**Silent prostatism**": asymptomatic renal failure from BOO. MTOPS: none in >3000 men. †Catheter first, watch post-obstructive diuresis, let creatinine stabilise, then surgery.
- **Bladder stones**: ≈ **8×** commoner, renal stones not (stasis).

- **Haematuria**: ↑ microvessel density; **finasteride** first-line (VEGF). †Exclude cancer and stones first.
- **UTI**: residual–UTI link unproven; MTOPS 0.1/100 patient-years. **Incontinence**: overflow or urge (DO in up to half).
- **Mortality** collapsed since 1950s; storage LUTS ↔ ↑ all-cause mortality. Delay harm: only indirect (lesser benefit of delayed TURP, VA).
- **Surgery** = soft end point (provider, era); AUR hard. Olmsted: moderate–severe symptoms, Qmax <12, volume >30, PSA ≥1.4 each ≈4× treatment; ≈1 in 4 treated in eighth decade. Baltimore: 0 risk factors → 3%, 3 → 37% surgery.

8 · Landmark trials

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure RR 0.48; 5-yr 10% vs 21%; ≈36% crossed over; delayed TURP smaller gains	Watchful waiting safe; delay may reduce benefit
PLESS	Finasteride vs placebo, 4 yr	Placebo returns to baseline; AUR/surgery rise with PSA/volume	PSA/volume predict progression
MTOPS	Placebo vs doxazosin vs finasteride vs combination	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4; no renal failure	Defined "clinical progression"
Olmsted County Study	Population cohort, 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural history; age, IPSS, Qmax, volume predict AUR

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

1 · Definitions and the diagnostic problem

- BPH = **histological** smooth muscle + epithelial proliferation in the **TZ**. Chain: BPH → BPE → BPO → BOO → LUTS.
- LUTS multifactorial: DO, DUA, stricture, UTI, metabolic, cardiovascular, neurological disease.

> **KEY POINT**
> Bothersome LUTS ≠ BPH-related BOO until assessed. Aims: **grade severity/bother** and **identify the cause**. Symptoms, flow, PVR, volume and cystoscopy cannot separate obstruction from bladder dysfunction; only the **pressure–flow study** can (gold standard, reserved for when it changes surgery).

Guidelines: AUA BPH (2021); EAU non-neurogenic male LUTS (2022).

2 · History and symptom assessment

- **Comorbidities**: diabetes mellitus/insipidus, renal disease, heart failure, sleep apnoea, neurological disease (Parkinson, MS, stroke, SCI, disc). Haematuria, prior lower-tract surgery.
- **Drugs**: loop diuretics, β-blockers worsen storage; anticholinergics impair contractility. Excess fluid/caffeine worsen storage.
- Assess **sexual function**. LUTS do **not** raise PCa risk (detection bias; PCPT placebo arm).
- Voiding: weak stream, hesitancy, straining, intermittency, incomplete emptying, dribbling. Storage: frequency, urgency, urge incontinence, nocturia (**OAB**).
- P3 IPSS score — *AUASI 7 items + QoL (0–6, WHO 1993), subscores not validated, does not diagnose BOO, VPSS/BII/OAB-q alternatives*
- AUASI/IPSS total **0–35**: mild **0–7**, moderate **8–19**, severe **20–35**.
- Minimum perceptible change **3 points**; with severe baseline, **percentage** reduction more reliable.

3 · Examination and DRE

- Suprapubic bladder (retention); penis: meatal stenosis, phimosis, urethral mass, discharge; small testes → low testosterone (metabolic syndrome); **anal tone**, perineal/lower-limb neurology.
- **DRE**: volume + exclude PCa (low sensitivity; with PSA reduced overdiagnosis in ERSPC). **Underestimates PV above 30 mL**; PPV good at 30 mL, near-perfect at 50 mL; flat gland → <30 mL.
- PV does not correlate with symptoms/obstruction, but **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · Frequency–volume charts and nocturia

- ICS: micturition chart (times only) → FVC (times + volumes, 24 h) → bladder diary (+ intake, incontinence, pads, urgency).

- **Indication**: predominant **storage LUTS or nocturia**; more accurate than recall. **Duration** **3 days** (adherence falls at 7). Has a training effect — keep routine.
- **Nocturia**: waking ≥1× to void. **Nocturnal polyuria**: night output **>20%** of 24 h (young), **>33%** over 65.

> **ALGORITHM**
> **Nocturia**: FVC → †global polyuria (>40 mL/kg/24 h → diabetes, intake) / nocturnal polyuria (heart failure, sleep apnoea, evening fluids, diuretic timing) / reduced capacity (OAB, BOO) → treat cause → persistent → consider **sleep apnoea** (AUA).

Nocturnal polyuria: no fluid 2 h before bed, avoid evening alcohol/caffeine, retime diuretics ± **desmopressin**. †Risk **hyponatraemia**: sodium at baseline, first week, 1 month.

5 · Laboratory tests

- **Urinalysis mandatory**: UTI, microhaematuria, glycosuria, proteinuria. Pyuria + nitrite insensitive in painless LUTS (reference 10^5 CFU/mL). Gross haematuria → exclude bladder cancer. **Cytology**: severe storage + dysuria, smokers (CIS).
- **PSA** — PCa risk, volume surrogate, progression predictor. Only if life expectancy **>10 years** and result changes management.
- PSA predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s); AUC ≈0.76–0.78. Baseline PSA predicts AUR, surgery, symptom/Qmax change.
- **5-ARIs ↓ PSA 40–50% by 12 months** → baseline first. †Double PSA after ≥6 months; any rise from nadir needs evaluation.
- **Creatinine/eGFR**: EAU if hydronephrosis, renal insufficiency or retention suspected; AUA not explicitly. Renal failure <1%/4 yr (MTOPS); raises surgical complications.

6 · Instrumental investigations

PVR

- Ultrasound soon after voiding (catheter = gold standard, invasive). No standard normal.
- **<30 mL** insignificant; persistently **>50 mL** important; **200–300 mL** may mean DUA, poorer response; **>300 mL** = significant chronic retention.
- High PVR = BOO **or DUA**. Predicts symptom deterioration (MTOPS >39 mL; ALTESS >93 mL), **not surgery**. Rising PVR predicted AUR; ≥500 mL predicted AUR recurrence.

Uroflowmetry

- **Qmax**, voided volume, pattern (average flow less accurate). Inaccurate if voided volume **<150 mL** → repeat; void in preferred position.
- **Cannot distinguish BOO from DUA** (DUA 9–48%). Qmax **10 mL/s** more specific; **15 mL/s** more sensitive.
- Optional initially; **recommended before active treatment** and for monitoring.

Urodynamics

- P2 Role of UDS before prostate surgery — *cystometry/PFS, BOOI and BCI, BOO = high pressure + low flow vs DUA, EAU indications, UPSTREAM: not routine*
- DO common in BPH; cystometry identifies antimuscarinic responders. DUA predicts long-term surgical failure; contractility independent of treatment/duration of obstruction.

- PFS indications (EAU): failed invasive treatment; cannot void **>150 mL**; PVR **>300 mL**; age **>80** or **<50 years** with voiding LUTS; neurological disease, pelvic surgery. AUA: before surgery if **Qmax >10 mL/s**.

Imaging and endoscopy

- P1 Methods to estimate prostate size — *TAUS/TRUS ellipsoid, planimetry, MRI, CT; IPP grading*
- Upper-tract ultrasound not routine; only for raised creatinine, large PVR, haematuria, UTI, stones. IVU obsolete.
- **TRUS** = method of choice for PV; needed before 5-ARI and **any BPH surgery**; median lobe guides MIST choice. PV predicts progression/AUR (ALTESS >49 mL) but weakly BOO.
- **IPP ≥10 mm** = high grade; predicts need for therapy and **failed TWOC**; not a PFS substitute. BWT (2 mm), DWT, UEBW, penile cuff, NIRS: accurate but **none replaces PFS**.
- **Cystoscopy** neither diagnoses BOO nor decides treatment. Indications: gross haematuria/bladder cancer history, recurrent UTI, stricture, prior surgery, **exclude median lobe before PUL**; young man with small prostate → **bladder-neck contracture**. VCUG: stricture, diverticulum, reflux.

7 · Evaluation algorithms

ALGORITHM

EAU (≥40 yr): history (incl. sexual) → symptom score → exam + DRE → urinalysis → PVR → PSA (if LE >10 yr). Abnormal DRE/PSA/urinalysis or neurological → relevant guideline. Not bothersome → no work-up. Significant PVR → renal US ± function. Bothersome → FVC (storage/nocturia) → prostate US + uroflow if treating → medical algorithm. Surgery → endoscopy if it alters choice; PFS for specific indications.

ALGORITHM

AUA: history, exam, IPSS, urinalysis ± PVR/flow; **PVR >300 mL → non-neurogenic chronic retention** → counsel (behavioural, medical, procedural) → reassess **4–12 weeks**: improved → follow-up; nocturia → FVC + drug; incomplete → FVC, PVR/flow, UDS, cystoscopy → OAB pathway, surgery, or both.

8 · Watchful waiting (WW)

- **Indication**: mild–moderate uncomplicated LUTS, minimal bother. **Excluded**: gross haematuria, UTI, raised PSA or PVR, neurological disease.
- Reassurance, education, lifestyle, monitoring; ≈half stable over 15 years. VA trial: most WW men did not fail; ≈quarter crossed over (high **bother**). Self-management sessions reduced failure. Review **yearly**.
- Lifestyle: fluids **1.5–2 L/day**, none 2 h before bed; less caffeine/evening alcohol; retime diuretics; **double voiding**, **urethral milking**; avoid constipation; bladder retraining (suppress urge, target ≥3 h or 200–400 mL).

KEY POINT

Treatment is driven by **bother and complicating features**, not score or size alone.

9 · Medical therapy — choice

- First line without **absolute surgical indication** (refractory/recurrent AUR, recurrent UTI, renal insufficiency, stones, recurrent haematuria *→ Ch 151*).

- P2 BPH: medical management and surgical modalities — *drug classes and doses, MTOPS/CombAT/SMART, IFIS, ALFAUR, desmopressin, surgical options by size*

ALGORITHM

EAU: not bothersome → WW | nocturnal polyuria → lifestyle ± desmopressin | storage → antimuscarinic or β3-agonist | **PV >40 mL, long-term** → 5-ARI ± α1-blocker/PDE5I | otherwise α1-blocker or PDE5I → residual storage → add antimuscarinic/β3-agonist.

AUA: α-blocker (PDE5I if ED) → **prostate >30 mL** → add 5-ARI → PDE5I trial → surgery.

KEY POINT

Reading trials: placebo improves scores and flow (4-week run-in ideal); judge by 3-point/percentage threshold; Qmax as %. Goals: relieve symptoms, reduce BOO, improve emptying, **prevent progression**.

10 · α1-Blockers

- P2 Tamsulosin; Silodosin. — *α1a ≈70% of prostatic α1, subtype table, dynamic component, onset, ejaculatory dysfunction, IFIS, silodosin renal dosing, greatest BOOI fall*
- Do **not** shrink the gland. α1d blockade may ↑ capacity, ↓ DO, improve bladder perfusion. α1b → vascular AEs; α1d on detrusor (twice α1a).

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; abandoned
Short-acting	Prazosin, indoramin	bd	Not used for LUTS
Long-acting	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin IR	2–8 mg od	Start 2 mg at bedtime, titrate
	Doxazosin GITS	4–8 mg od	Start **4 mg**, **no titration**
Uroselective	Alfuzosin ER	10 mg od	Clinically uroselective
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d
	Silodosin	8 mg od	Most α1a-selective
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia

KEY POINT

Titration: α1b blockade → **first-dose hypotension**/syncope; bedtime start + titration avoid it. Uroselectives fixed-dose; doxazosin GITS starts 4 mg untitrated.

- Level 1a; ≈double placebo symptom gain, modest Qmax; **durable** (PREDICT, MTOPS). Tamsulosin: flow in hours, symptoms by 1 week.
- Tamsulosin **0.4 mg** plateau (0.8 adds AEs); alfuzosin 15 = 10 mg; terazosin best at 10 mg. Naftopidil more storage relief.
- **No α1-blocker clearly more effective**; they differ in AEs. Doxazosin ↓ progression 39% (MTOPS), **not AUR/surgery**.
- **AEs**: dizziness, asthenia, orthostatic hypotension (standing systolic fall ≥20 mm Hg + symptoms), anejaculation. BP fall significant only in uncontrolled hypertensives.
- **ALLHAT**: doxazosin arm stopped (heart failure) — not sole antihypertensive; **never two α-blockers**.
- **Elderly**: falls/fractures; severe hypotension mostly in **first 8 weeks** — check BP in week 1.

- **Sexual**: ED = placebo. Ejaculatory dysfunction: retrograde (bladder neck), **anejaculation** (emission failure), central. Silodosin (OR ≈32) > tamsulosin (OR ≈9) > alfuzosin; younger men more affected.

> **KEY POINT**
> **IFIS**: **billowing iris, iris prolapse, progressive miosis**. α1a blockade → dilator atrophy → occurs **years after stopping**; stopping pre-op does **not** help. **Tamsulosin** highest, then alfuzosin, doxazosin, naftopidil. †Awaiting cataract surgery → defer α1-blocker.

Drug	Vascular	Ejaculatory	IFIS
Terazosin	High	Low	Reported
Doxazosin	High (GITS lower)	Low	Lower
Alfuzosin	Intermediate	Lowest uroselective	Intermediate
Tamsulosin	Low	Dose-dependent	**Highest**
Silodosin	Lowest	**Highest**	Not well studied

> **KEY POINT**
> One efficacy, three toxicities: **α1b → vascular** (terazosin, doxazosin); **α1a → ejaculation, IFIS** (silodosin, tamsulosin); **α1d → storage benefit** (naftopidil).

11 · 5α-Reductase inhibitors

- P2 Dutasteride. — *type 1+2, DHT ↓90%, 0.5 mg, PV ↓25%, AUR ↓57%/surgery ↓48%, PSA doubling, REDUCE, teratogenic*
- Relieve **static component**; leave HPG axis intact — only androgen class in use (castration, antiandrogens, GnRH analogues, aromatase inhibitors: inefficacious/toxic).
- **Finasteride 5 mg** (type 2; DHT †~70%; PV ↓18–20%; PLESS AUR ↓57%, surgery ↓55%). Head to head with dutasteride: no difference.
- **Slow**: maximum shrinkage at **6 months**; benefit needs ≥6 months.
- **Volume dependent**: best at **>40 mL** (PLESS PV >55 mL → 70% reduction). Indication: EAU >40 mL, AUA >30 mL, moderate–severe LUTS.
- **Only class altering natural history** (↓ AUR, surgery). PSA halves; 5-ARIs **improve PSA accuracy** for PCa.
- PCPT/REDUCE (23% ↓): fewer cancers, slight ↑ Gleason ≥8 → **not licensed for prevention** (possibly volume artefact). Finasteride ↓ refractory BPH haematuria.
- **AEs**: ED, ↓ libido, ↓ ejaculate (≈2× placebo; wane after 1–2 yr; ↓ cavernosal NOS). **Depression** (FDA 2011), self-harm. Gynaecomastia rare. **Post-finasteride syndrome**: persistent symptoms after stopping.

> **KEY POINT**
> 5-ARIs treat the gland: slow, volume-dependent, modest for symptoms, but prevent AUR and surgery — and halve PSA (trap and tool).

12 · Antimuscarinics

- Block detrusor **M2/M3**, mainly on **afferent** pathways.
- **Indication**: predominant storage LUTS, **PVR <200 mL**, no prior AUR.

- Agents: darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine ER 4 mg, trospium.
- Monotherapy: storage relief, no total IPSS/Qmax change; favoured by small prostate, PSA <1.3, higher Qmax. In BOO: small PVR rise, AUR = placebo (trials excluded PVR ≥200).
- AEs: **dry mouth**, constipation, cognitive impairment. **Oxybutynin** most CNS penetration → avoid in elderly; trospium least†.

> **KEY POINT**
> Antimuscarinics treat the bladder, so every recommendation is anchored to a PVR ceiling (200 mL).

13 · β3-Agonists

- β3 → cAMP → ↓ calcium → **storage relaxation without weakening voiding** (Qmax unchanged).
- **Mirabegron** **25 mg** → **50 mg**; BOO safety trial: PVR rose only at 100 mg; less dry mouth.
- AEs: nasopharyngitis, headache, BP ≤1 mm Hg. **Contraindicated** in uncontrolled hypertension **≥180/110 mm Hg**; QT at 200 mg; moderate CYP2D6 inhibitor†.
- **Vibegron** 75 mg: half-life 25–38 h, no CYP2D6 inhibition; data mainly in women (EMPOWUR).

14 · PDE5 inhibitors

- NO–cGMP: ↑ perfusion, smooth-muscle relaxation, antiproliferative, **afferent inhibition**, anti-inflammatory.
- **Tadalafil 5 mg daily** only licensed (2.5 mg ineffective). IPSS ↓ ≈6 (placebo 3.6), **independent of ED**; ↓ nocturia. **Qmax unchanged**.
- First choice LUTS + ED (AUA). AEs: headache, back pain, dyspepsia, flushing.

†Contraindicated with nitrates, riociguat.

15 · Combination therapy

- Short trials (6–12 mo, glands <40 mL) showed no benefit.
- **MTOPS**: progression = AUASI rise **≥4**, AUR, UTI/urosepsis, incontinence, creatinine rise ≥1.5 mg/dL. Placebo 17%. NNT combination 8.4 (≈5 if PSA >4 and PV >40).
- **CombAT** (PV ≥30, PSA ≥1.5): AUR/surgery ↓ two-thirds vs tamsulosin, NS vs dutasteride.
- **Indication**: high risk — PV **>30–40 mL**, PSA **>1.5 ng/mL**. **SMART**: stop α-blocker after 6 months.

MTOPS ↓ vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

> **KEY POINT**
> **α-blockers prevent symptomatic progression only; 5-ARIs prevent AUR and surgery.**
> α-blocker = fast dynamic relief; 5-ARI = slow static shrinkage.

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS (tolterodine 4-arm; NEPTUNE)	PVR <150–200 mL only; AUR <1–3%
α1-blocker + β3-agonist	Persistent storage (MATCH)	PVR rise; limited data
α1-blocker + PDE5I	LUTS ± ED; small gain	Hypotension; not recommended; start lowest dose
PDE5I + 5-ARI	Large prostate ± ED	Single trial; FDA capsule 2021

16 · Phytotherapy

- **Every extract unique** (product-specific results). Serenoa does not lower PSA.
- **Serenoa repens**: large trials (ethanolic/CO_2) no benefit. **Hexane extract (Permixon) 320 mg daily with food**: fewer nocturnal voids, ↑ Qmax — EAU option to avoid drug AEs. Rare warfarin interaction.
- Pygeum: modest; others unreliable. AUA/EAU: **no general recommendation**.

17 · Follow-up

- Progression risk: higher PSA, PV, PVR, age, bother; dynamic AUR predictors: worsening symptoms, rising PVR. CONDUCT: bother predicts benefit.

ALGORITHM

4–12 weeks (α-blocker response, AEs) → **6 months** (5-ARI) → **annually**: score, flow, PVR; PSA on 5-ARI if LE >10 yr. WW: yearly.

18 · Acute urinary retention

- P2 Catheter free trial — *spontaneous vs precipitated AUR, α-blocker before TWOC (ALFAUR), predictors of failure, recurrence, 5-ARI prevention*
- Spontaneous (~two-thirds; worse TWOC) vs precipitated. Incidence 2.2–8.5/1000 person-years.

ALGORITHM

Catheter (†suprapubic if fails) → **α1-blocker** → **TWOC ~3–5 days** → voids: continue (+5-ARI if large) → fails: up to **3 TWOCs** → recurrent/failure → PFS and surgery → *Ch 151*.

- Emergency surgery more morbid → TWOC first; success ≈ **60%**; recurrence mostly within **6 months**.
- Failure predictors: age, severe LUTS, drained **>1 L**, spontaneous AUR, high-grade IPP.
- Prevention: 5-ARIs (PLESS, dutasteride, MTOPS), not α-blockers (ALTESS, MTOPS). CombAT AUR ↓68% vs tamsulosin, 18% (NS) vs dutasteride.

KEY POINT

α-blocker rescues the TWOC (dynamic); **5-ARI prevents the next episode** (static).

19 · Landmark trials

Trial	Design	Result	Take-home
VA TURP vs WW	RCT, moderate LUTS	TURP better; most WW did not fail; crossover by bother	WW safe
PLESS	Finasteride vs placebo, 4 yr, PV ~55 mL	AUR 7→3%, surgery 10→5%; combined ↓51%	5-ARI prevents AUR/surgery
Dutasteride phase III	2 yr, PV ≥30 mL	DHT ↓90%, PV ↓26%, AUR ↓57%, surgery ↓48%	≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 yr	Progression ↓39/34/66%; AUR/surgery only with 5-ARI	Combination for high risk
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 yr	AUR/surgery ↓66% vs tamsulosin	Combination for large gland
SMART	α-blocker withdrawal	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI prevention	Fewer cancers, more high grade	Not licensed
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

1 · Scope and principles

- **AUA 2023 operations**: TUIP, TURP, TUVP, PVP, prostatic urethral lift (PUL; UroLift), water vapour thermal therapy (WVTT; Rezūm), temporary implanted prostatic device (TIPD; iTind), robotic waterjet treatment (RWT; Aquablation), HoLEP/ThuLEP, simple prostatectomy (open, laparoscopic, robotic), prostate artery embolisation (PAE).
- TUMT removed 2023 (legacy). Sham effect is large (sham endoscopy improves IPSS and Qmax) → new devices need sham-controlled RCTs.

KEY POINT

Volume predicts progression (AUR, surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** Gland >30 cc triples AUR risk (Olmsted County).

2 · Work-up and indications for surgery

- History, examination, IPSS, urinalysis at baseline (St 1); repeat IPSS after treatment (St 3). Bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery: PVR **required** (St 6); **consider** size/shape by TRUS, abdominal US, cystoscopy, MRI or CT (St 5), uroflowmetry (St 7), pressure–flow if diagnosis uncertain (St 8). Cystoscopy shows median lobe, stricture, tumour but is unreliable for size.
- Raised PSA → biopsy **before** BPH surgery. Counsel on failure/retreatment (St 9).

AUA Statement 26 (surgery recommended)	† EAU — absolute vs relative
Renal insufficiency from BPH Refractory retention Recurrent UTIs Recurrent bladder stones or gross haematuria from BPH LUTS refractory to, or unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation/renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5-ARI **Relative**: moderate–severe bothersome LUTS failing/declining medical therapy

- **Antibiotics (AUA 2019)**: single dose for most transurethral cases; cover gram-negatives and enterococci (1st/2nd-generation cephalosporin or co-trimoxazole), guided by antibiogram and urine culture. **Active symptomatic UTI → defer surgery**.

3 · Choosing the operation

Volume / situation	AUA 2023 options	Restrictions
<30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes
>80–150 cc / >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc

Volume / situation	AUA 2023 options	Restrictions
Higher bleeding risk	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Preserve erection and ejaculation	PUL or WVTT (St 37)	Within 30–80 cc

KEY POINT

Only enucleation spans every size band; only simple prostatectomy joins it above 80 cc. TUIP alone is confined to small glands. MISTs and RWT live in 30–80 cc — so volume and median lobe assessment precede every recommendation.

4 · Monopolar TURP (m-TURP) — the benchmark

- P1 P2 TURP instruments and techniques; Measures to decrease complications of TURP — *resectoscope, Nesbit/floor-first resection, canoe chips, prevention of each complication*
- AUA: TURP for <30 and 30–80 cc (St 28); monopolar or bipolar by expertise (St 29). Monopolar irrigant non-ionic (water, **1.5% glycine**†, sorbitol) and **hypo-osmolar** → TUR syndrome.
- Grounding pad away from joint prosthesis.
- Over-resecting dorsal bladder neck → scope goes "uphill", trigone undermined.

ALGORITHM

Inspect bladder (tumour → resect it, defer TURP) → orifices, neck, trigone, veru, sphincter (indigo carmine/methylene blue if orifices hidden) → median lobe → lateral lobes → anterior and apex last → evacuate chips → haemostasis → large three-way catheter

- **Anterior/apex last**: anterior fossa has least adenoma and perforates easily; sphincter tilted (anterior part more proximal).
- **Arterial**: bright, persists on filling and draining; bladder-neck bleeders seen best with bladder empty; always fulgurate (loop tamponade first; "ricochet" bleeders may need deeper resection). **Venous**: darker, vanishes with full bladder; after arteries secured → balloon 50–60 mL + traction.
- Post-op: even routine TURP absorbs **800–1000 mL** irrigant. Catheter out 24–48 h; stool softeners 2–4 weeks; no cycling 4–6 weeks; no sex ≥2 weeks.
- Warn: dysuria for weeks; eschar bleeding at **1–4 weeks**; storage symptoms → antimuscarinic/β3-agonist once emptying confirmed (detrusor overactivity may take months). Outcomes durable ~10 years.

5 · TURP complications

- P1 P3 TUR Syndrome — *mechanism, glycine toxicity, features, risk factors, 3% saline protocol*
- Risk: gland **>45 cc**, resection **>90 min**, perforation, irrigant height (keep ≈ **60 cm**; venous pressure ≈ **10 mm Hg**). Severe: Na^+ **<120 mEq/L**.

ALGORITHM

Suspect → stop/expedite, haemostasis, stop irrigation → Na^+ → †loop diuretic if overloaded → severe CNS: **3% saline 100 mL** (+2–3 mEq/L), repeat ×2 at 10 min (max 300 mL); target +4–6 mEq/L; ≤12 mEq/L in 24 h (≤8–10 if presentation >48 h) → HDU

KEY POINT
Acute post-TURP hyponatraemia is safer to correct than chronic: brain has not yet shed organic osmolytes, so osmotic demyelination less likely. Bipolar saline TURP eliminates TUR syndrome.

- **Perforation** (glistening fat): extraperitoneal limited → prolonged catheter; extensive → percutaneous/open drainage; dome → cystogram; intraperitoneal → laparotomy.
- **Ureteric orifice**: 70° lens, IV dye; large median lobe → resect from midline first. **Intra-op erection**: wait, else intracorporeal phenylephrine.

Late complication	Cause / features	Management
BNC	Over-resection/fulguration (small glands); onset ~6 months, flow declines	Dilation → incision (Collins knife/urethrotome/laser) over wire into perivesical fat → recurrent: + mitomycin C (10 mL of 0.4 mg/mL) → intractable: Y-V plasty. Obliterated: IV methylene blue + suprapubic pressure
Stricture	Meatus/fossa navicularis	Dilate meatus 2 Fr above sheath; 30 mL gel
Retrograde ejaculation	About **two-thirds**	Counsel; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost; apical over-resection → total SUI; rare, usually transient	Veru = distal limit; †pelvic floor → †sling/AUS
Bleeding	Early: incomplete haemostasis; delayed 1–4 weeks (eschar)	Early major → theatre; delayed mild → fluids, †clot evacuation if retention
ED	No conclusive causal evidence	—
Retreatment	Residual apical/anterior tissue, >80 cc, regrowth, BNC, stricture	Rises with size and pre-op retention

6 · Bipolar TURP (b-TURP)

- P2 Monopolar v/s Bipolar TURP — *saline, no TUR syndrome (NNT ≈50), less clot retention (NNT ≈20), equal efficacy to 5 y*
- Equally effective and safer — likely to replace m-TURP as gold standard.

7 · TUIP

- Releases capsule and bladder neck. **Only <30 cc** (St 31), no median lobe (relative CI; EAU same). Ideal: young man, small gland, wants ejaculation/fertility.

ALGORITHM
Collins knife / loop / Ho–Tm laser → 5 and/or 7 o'clock incision from distal to orifice, through bladder neck, to proximal to veru → deepen to capsule (± fat) → cautery → catheter out, home day 0–1

- vs TURP: shorter, less bleeding, similar symptom relief, smaller Qmax gain, **more reoperation**, less retrograde ejaculation, no BNC.

8 · Bipolar TUVP

- Broad electrode (PlasmaButton) vaporises + coagulates → excellent haemostasis; no new capital; **no specimen**. Only **bipolar** recommended (St 32: <30 and 30–80 cc); monopolar abandoned (10 mm coagulation → irritation, SUI, poor durability).
- vs TURP: shorter catheter, fewer transfusions; more dysuria, AUR, retreatment; retrograde ejaculation as common.

> **KEY POINT**
> Vaporisation trades haemostasis for a coagulated, specimen-free bed: coagulum → dysuria and retention; no anatomical plane → weaker durability. Same trade with PVP.

9 · Laser physics

- P3 Lasers in BPH, including HoLEP — *wavelength/chromophore/penetration table (thulium **0.2 mm** shallowest; Nd:YAG abandoned)*
- Diode: †940–1470 nm, †water + haemoglobin, 0.5–5 mm; needs comparative data. Safety: signage, covered windows, eyewear for patient too.

10 · PVP (GreenLight)

- P4 Green light lasers — *physics, technique, AUA <80 cc + bleeding risk, GOLIATH, complications*
- No bubbles = charring → dysuria. Posterior tissue in horizontal layers, never at rectum; avoid heavy coagulation at neck (BNC).
- Bleeding: moderate → vaporise around feeder; pulsatile → coag ~30 W; refractory → Bugbee (non-ionic irrigant) or resection.
- Tissue passage 7–10 days. Dysuria ≤3 months → NSAIDs or steroid taper; exclude flap/retained fibre fragment. Infection → longer culture-directed antibiotics.
- Anticoagulated men improve equally without more transfusion.

11 · HoLEP

- P4 P1 HoLEP — *size-independent (St 38/43), only endoscopic method superior to TURP; settings, steps, learning curve 20/50, BNC if <40 cc*

> **ALGORITHM**
> 5 and 7 o'clock bladder-neck incisions (6 o'clock if no median lobe) → to veru, enucleate median lobe → perimontanal inverted-U, apical plane → sweep anteriorly to 11–1 o'clock → incise apical mucosal strip (no sphincter traction) → anterior plane to neck ("Baleen" sign = bladder) → lobes into bladder → defocused haemostasis → morcellation via nephroscope → 22 Fr three-way, ~60 mL balloon

- Smooth reflective capsule = depth reference: too shallow leaves adenoma/feeders; too deep perforates (few days' catheter).
- Morcellator: mucosa engaged → stop cutting and suction. Retained adenoma does not dissolve → re-morcellate.
- **Transient SUI** common early (∝ volume removed); early apical release reduces it; careless commissure splitting → permanent SUI.
- Retrograde ejaculation ≈ **three-quarters**; median-lobe-only ~90% preserved. Erections unchanged/improved. Retreatment rare (whole TZ removed, unlike TURP's peripheral rim); durable to 18 years. Anticoagulated: slightly longer LOS/CBI; warfarin worse than DOACs.

12 · Thulium (ThuLEP) and variants

- Continuous wave may carbonise. ThuVARP (vaporesection), ThuVAP, ThuVEP (vapo-incision), ThuLEP (mostly blunt dissection). Same steps/morcellation as HoLEP.
- ≈ HoLEP function; less bleeding than TURP; St 38 and 43. Retrograde ejaculation ~four-fifths; erections preserved.

	HoLEP	ThuLEP / ThuVEP
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MISTs

Ideal MIST: office-based, single session, no GA, spares sexual function, ends medication. **Trade-off**: retreatment within 5 years exceeds surgery; durability beyond 5 years uncertain.

- P4 UroLift (prostatic urethral lift) — *implant, 30–80 mL no median lobe, >1.5 cm from neck, L.I.F.T./BPH6, encrustation*
- PUL: typically **4–6 implants** anterolaterally between dorsal veins and NVBs; FDA ≤100 cc not adopted by AUA. Encrusted implants → endoscopic graspers.
- **WVTT (Rezūm)**: **convective** — RF steam into TZ, condensation releases latent heat (540 cal/g) → 1.5–2 cm necrosis; capsule collagen confines heat; ~4 kJ total.
- 9-s injections at 103 °C, first 1 cm distal to neck at 3 and 9 o'clock, then every 1 cm to veru; extra for **median lobe**; sedation/prostatic block; ~5 min. Never aim straight posterior near veru — angle 45° off midline.
- **Rezūm II**: ~50% IPSS gain to 5 years; retreatment 4.4% (underestimated); **no de novo ED**; reduced ejaculate settles by 12 months. Catheter few days; irritation ~3 weeks. St 36/37.

> **KEY POINT**
> PUL moves tissue; WVTT ablates it. Hence PUL cannot treat a median lobe, needs no catheter, minimal dysuria; WVTT treats median lobes at the cost of catheter and weeks of irritation. Both preserve bladder neck → antegrade ejaculation.

14 · Robotic waterjet (RWT, Aquablation)

- Non-thermal saline waterjet; resection mapped on real-time biplanar TRUS; robot ablates in minutes (up to 2.4 cm deep); neck, veru, sphincter spared. **Needs GA/spinal — not a MIST**. 30–80 cc (St 39); median lobe treatable; no specimen; mainly 4–8 o'clock tissue.
- **Haemostasis is the weakness**: traction → bleeding/transfusion; now **focal bladder-neck cautery** via resectoscope.
- **WATER**: non-inferior to TURP to 5 years; retrograde ejaculation 10% vs 36%. Rectal injury rare, catastrophic.

15 · True MISTs and temporary devices

- True MISTs: flexible cystoscopy, no sedation; durability 12–18 months by design.
- **iTind (TIPD)**: three nitinol struts (12, 5, 7 o'clock) + 6 o'clock anchor; left **5–7 days** → ischaemic remodelling; removed via 18 Fr catheter. St 41: 25–75 cc, no obstructing median lobe. Sham RCT: responder end point met, absolute IPSS difference not significant.

- **Optilume**: anterior commissurotomy + paclitaxel balloon (≥5 min); single-arm data; not in AUA list.

16 · Prostate artery embolisation (PAE)

ALGORITHM

Femoral access, local anaesthesia → pelvic angiography → catheter balloon as landmark → 1.9–2.4 Fr microcatheter into prostatic arteries (mostly from internal pudendal; often duplicated) → microspheres to stasis — bilateral better

- **Contraindications**: underactive/acontractile or neurogenic bladder, stricture/BNC, large diverticula/stones, contrast allergy, CKD. Atherosclerosis/tortuosity may prevent completion; high radiation.
- AUA St 40: trained clinicians; no size limit; role in refractory BPH haematuria. EAU: men accepting inferior results to TURP.
- vs TURP: inferior IPSS, Qmax, volume reduction; shorter stay; most reintervene by 10 years.
- Complications: access haematoma/pseudoaneurysm; non-target embolisation (bladder, rectum, penis; usually conservative); post-embolisation syndrome, AUR.

17 · Overall comparison

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/GA; 1–2 d	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low
TUIP	<30 cc	Day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritation	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT

"Gold standard" for a given man = operation meeting his goals safely: shared decision on bothersome symptoms, size/shape, bleeding risk, ejaculation wishes, surgeon expertise.

18 · Landmark trials

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	Lesser-efficacy option

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

1 · Definition and surgical anatomy

- Enucleation of BPH adenoma only; gold standard for outlet obstruction with glands **>80 g**.
- Plane: **surgical capsule (pseudocapsule)** = peripheral zone compressed by enlarging transition zone → *Ch 148*.
- **Urethral arteries enter posterolaterally just distal to bladder neck → 5 and 7 o'clock** figure-of-eight. Dorsal venous complex (DVC) → internal iliac.
- Eponyms MCQ: **Freyer** suprapubic transvesical; **Millin (1945)** retropubic; first laparoscopic (Mariano) 2002, robotic (Sotelo) 2008.

2 · Indications

- Gland **>80–100 g** with bothersome LUTS refractory to medical/endoscopic therapy.
- Retention, recurrent UTI, **bladder stones**, haematuria, obstructive nephropathy.
- Diverticulum needing surgery (relative); cannot be placed in lithotomy for TURP.

3 · Pre-operative evaluation

- Beyond PSA/MRI, TRUS volume, cystoscopy, urodynamics (atonic bladder after long retention?): bleeding history/anticoagulants, prior pelvic surgery (adhesions dictate route), **urine culture** (treat first), coagulation.
- General anaesthesia preferred; spinal acceptable for open only.

> **KEY POINT**
> Work-up asks: big enough (TRUS)? cancer (DRE, PSA, MRI)? will the bladder empty (urodynamics)?

4 · Open techniques

- P1 P2 Millin's prostatectomy — *Retzius, transverse capsulotomy ~2 cm distal to bladder neck, bladder not opened, apex divided under vision, 5/7 o'clock sutures, watertight closure; Millin vs Freyer table*
- **Freyer**: transverse **cystotomy 1–2 cm proximal to bladder neck**; mucosal incision round neck avoiding orifices; finger fractures anterior commissure → enucleate to apex. Treats **median lobe, diverticula, stones**; apex blind. Two-layer closure ± Malecot SPT.
- **Millin** extra: endopelvic fascia, puboprostatic ligaments divided; superficial DVC cauterised. Drawback **DVC bleeding**.
- **Vesicocapsular** (vertical across neck and capsule): more leak, sphincter risk. Catheter 30-mL balloon inflated to 50 mL (cannot slip into fossa).

> **KEY POINT**
> All open variants = enter → pseudocapsular plane → divide urethra at apex **proximal to verumontanum**, no sphincter traction → 5/7 o'clock haemostasis + bladder neck mucosal advancement.

5 · Haemostasis

Technique	When / how
Pack fossa 5 min	First
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule
Bladder neck mucosal advancement ("trigonisation")	Haemostasis; prevents contracture
11 and 1 o'clock sutures	Anterior branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC and lateral pedicles, sparing NVBs
O'Conor plication	Diffuse bleeding: transverse sutures approximate posterior capsule
Malament purse-string	Persistent bleeding: nylon purse-string round neck, tied suprapubically over a button → partitions fossa from bladder; out day 2–3

ALGORITHM

Postoperative bleeding → catheter traction (50 mL balloon) ± CBI → exclude coagulopathy → endoscopic clot evacuation + fulguration → reopen fossa if severe

6 · Postoperative care (open)

- Day 2, clear urine: Foley out, clamp SPT for trial of void. High drain output → **drain creatinine**. No heavy lifting ≈4 weeks.

7 · Laparoscopic and robotic

- P3 Laparoscopic Trans vesical Prostatectomy — *steps, same function as open with less morbidity*
- **Multiport RASP**: usually transperitoneal; extraperitoneal for hostile abdomen.

ALGORITHM

Vertical cystotomy → orifices → bladder neck incision 3→9 o'clock under median lobe → **posterior plane** to apex → lobes → anterior commissure, see veru → transect urethra **just proximal to veru** → 5/7 barbed sutures → **vesicourethral mucosal advancement** (double-armed from 6 o'clock) → two-layer closure

- Avoid posterolateral cautery (NVB → ED). Mucosal advancement → usually **no CBI, drain or SPT**; home day 1, catheter out day 5–7.
- **Single-port transvesical (STEP)**: 2.7-cm incision, extraperitoneal, **pneumovesicum 8 mm Hg** → same-day discharge, catheter out day 2.

8 · Outcomes and comparisons

Comparison	Bottom line
Robotic vs open	Less bleeding, shorter catheter/stay, fewer complications; longer operation. Function **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, **no TUR syndrome**, treats stones/diverticula; but more bleeding, longer stay

Comparison	Bottom line
†vs HoLEP	†RCTs, large glands: equivalent durable outcomes; HoLEP less bleeding, shorter stay; long learning curve, no diverticula → *Ch 151*

KEY POINT

Minimally invasive routes improve perioperative morbidity only; function identical because enucleation identical. Open transfusion now **<5%**.

9 · Complications

Complication	Cause / rate	Management
Extravasation	Incomplete closure; capsular tear	Drain creatinine = urine → keep drain + Foley
Ureteric orifice injury	Neck incision/sutures	Stent
Bladder neck contracture	Open **2–6%**; minimally invasive <1%	Dilation or Collings knife incision to 22 Fr
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare	Bulking (mild); AUS (severe)
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op
UTI, stricture, VTE	VTE <1%	Prophylaxis

KEY POINT

Adenoma lies inside pseudocapsule, proximal to veru: bladder neck lost (retrograde ejaculation); sphincter and NVBs spared (incontinence, ED rare).

★ Putting it together: the BPH pathway

A · Man with LUTS

ALGORITHM

1. Assess: IPSS (0–7 mild, 8–19 moderate, 20–35 severe), DRE, urinalysis, PSA if management changes, voiding diary if nocturia; uroflow + PVR before intervention → *Ch 150*.

2. Complications → surgery: refractory retention, recurrent UTI, stones, prostatic haematuria, renal impairment.

3. Mild → watchful waiting. **4. Bothersome** → drugs (B); failure → procedure (C).

B · Medical therapy

Drug	Use	Notes
α1-blocker	First line; days	Dizziness, ejaculatory dysfunction; floppy iris (cataract surgery)
5-ARI	Prostate >30–40 mL, long-term	Slow; less retention/surgery; PSA ↓40–50% (double)
α-blocker + 5-ARI	Large gland, progression risk	Best (MTOPS, CombAT)
Add antimuscarinic/β3-agonist	Persistent storage LUTS	Check PVR first
Tadalafil 5 mg daily	LUTS + ED	No nitrates
Desmopressin	Nocturnal polyuria	Sodium, esp. ≥65

C · Choosing a procedure

Situation	Options
<30 mL	TUIP or TURP
30–80 mL	Bipolar TURP, PVP, HoLEP/ThuLEP, Aquablation; MISTs (urethral lift, water vapour)
>80 mL	HoLEP/ThuLEP; simple prostatectomy (stones/diverticulum) → *Ch 152*
Anticoagulation continued	HoLEP, ThuLEP, PVP
Preserve ejaculation	Urethral lift (no median lobe) or water vapour
Unfit	Long-term catheter or †prostatic artery embolisation

D · TURP complications

KEY POINT

TUR syndrome (glycine absorption): confusion, visual change, hyponatraemia → stop, check sodium, diuretic if overloaded, 3% saline if severe, correct slowly; bipolar prevents. Late: retrograde ejaculation (~two-thirds), stricture (meatus, bulb), bladder neck contracture (small gland, over-resection; ~6 months), incontinence (pelvic floor → sling/AUS) → *Ch 151*.